

Solⁿ (Slide no: 17 and 19)

Solⁿ (Slide: 17) $f(x,y) = \frac{x+y}{30}$, $x=0,1,2,3$
 $y=0,1,2$

(I)

x	y	0	1	2
0		0	$\frac{1}{30}$	$\frac{1}{15}$
1		$\frac{1}{30}$	$\frac{1}{15}$	$\frac{1}{10}$
2		$\frac{1}{15}$	$\frac{1}{10}$	$\frac{2}{15}$
3		$\frac{1}{10}$	$\frac{2}{15}$	$\frac{1}{6}$

(i) $P(X \leq 2, Y=1)$
 $\downarrow \quad \quad \downarrow$
0, 1, or 2 1

Combination (0,1), (1,1) & (2,1)

$$\Rightarrow P(X \leq 2, Y=1) = f(0,1) + f(1,1) + f(2,1) \\ = \frac{1}{30} + \frac{1}{15} + \frac{1}{10} = \frac{1}{5}$$

(ii) $P(X > 2, Y \leq 1)$
 $\downarrow \quad \quad \downarrow$
3 0 or 1

$$= f(3,0) + f(3,1) = \frac{1}{10} + \frac{2}{15} = \frac{7}{30}$$

(iii) $P(X > Y)$

$$= f(1,0) + f(2,0) + f(2,1) + f(3,0) + f(3,1) + f(3,2)$$

$$= \frac{1}{30} + \frac{2}{15} + \frac{1}{10} + \frac{1}{10} + \frac{2}{15} + \frac{1}{6} = \left(\frac{3}{5} \right)$$

$$(iv) P(X+Y \leq 4)$$

$$\begin{array}{cc} \downarrow & \downarrow \\ 2 & 2 \\ 3 & 1 \end{array}$$

$$= f(2,2) + f(3,1) = \frac{2}{15} + \frac{2}{15} = \frac{4}{15}$$

Solution: (Slide 19)

$$f(x,y) = \begin{cases} 4xy, & 0 < x < 1 \\ & 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

$$(i) P(0 \leq x \leq \frac{1}{2}, \frac{1}{4} \leq y \leq \frac{1}{2})$$

$$= \int_{x=0}^{\frac{1}{2}} \int_{y=\frac{1}{4}}^{\frac{1}{2}} 4xy \, dy \, dx$$

$$= \int_{x=0}^{\frac{1}{2}} 4x \left(\int_{y=\frac{1}{4}}^{\frac{1}{2}} y \, dy \right) dx$$

$$= \int_{x=0}^{\frac{1}{2}} 4x \left(\frac{y^2}{2} \right)_{\frac{1}{4}}^{\frac{1}{2}} dx$$

$$= \int_{x=0}^{\frac{1}{2}} \frac{3}{8} x \, dx = \frac{3}{64}$$

$$ii) P(X < Y)$$

Since $X < Y$ i.e. $x < y$

given $0 < x < 1$ & $0 < y < 1$

$$\Rightarrow \boxed{0 < x < y}$$

we will take limit of 'x' is

limit of 'y' is given $0 < y < 1$

$$\Rightarrow P(X < Y) = \int_{y=0}^1 \int_{x=0}^y 4xy \, dx \, dy$$

$$= \int_{y=0}^1 4y \left(\int_{x=0}^y x \, dx \right) dy$$

$$= \int_{y=0}^1 2y^3 \, dy = \left(\frac{1}{2} \right)$$

—————x—————x—————